

Some commonly used surfaces

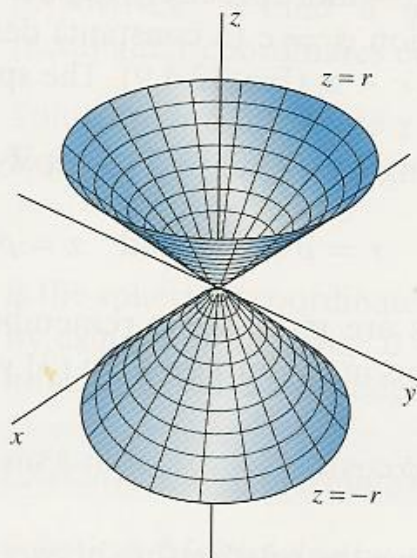


FIGURE 12.4.4 The cone $z^2 = r^2$.

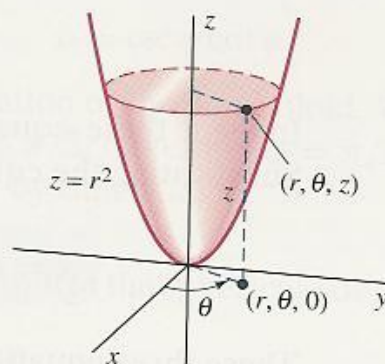


FIGURE 12.4.5 The paraboloid $z = r^2$.

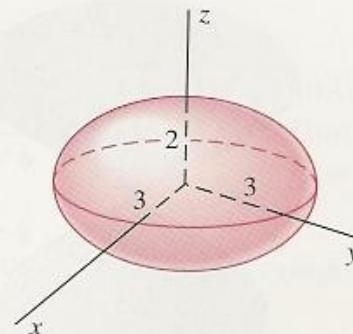


FIGURE 12.4.6 The ellipsoid $\frac{r^2}{9} + \frac{z^2}{4} = 1$.

Linear approximation

V EXAMPLE 1 Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

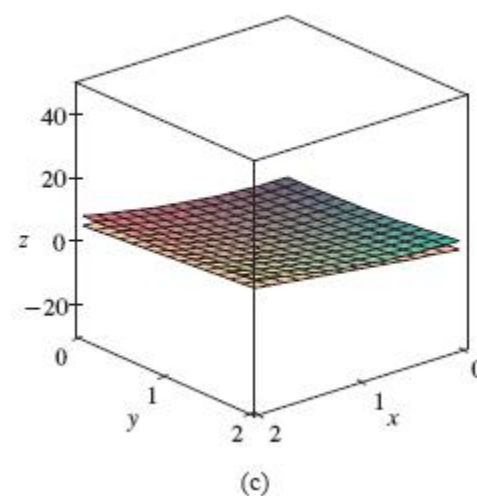
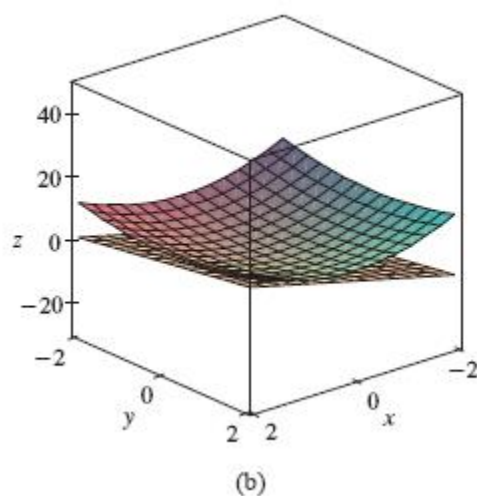
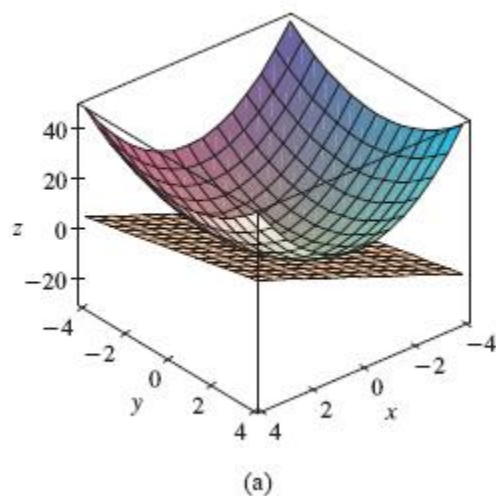
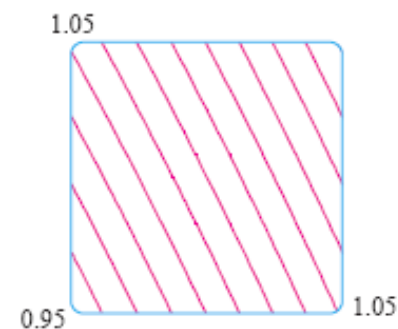
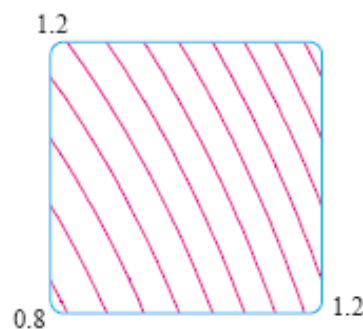
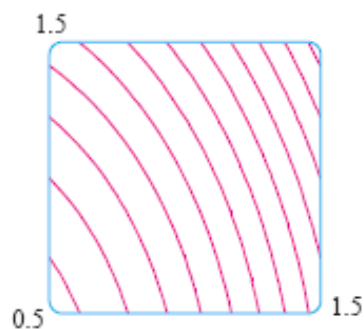
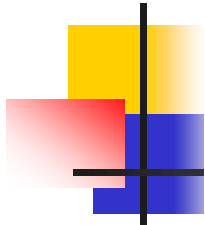


FIGURE 3

Zooming in toward $(1, 1)$
on a contour map of
 $f(x, y) = 2x^2 + y^2$



Equation of the tangent plane to the graph of the function $f(x, y) = 2x^2 + y^2$ at the point $(1, 1, 3)$ is $z = 4x + 2y - 3$.



The linear function of two variables $L(x, y) = 4x + 2y - 3$

is a good approximation to $f(x, y)$ when (x, y) is near $(1, 1)$. The function L is called the *linearization* of f at $(1, 1)$ and the approximation

$$f(x, y) \approx 4x + 2y - 3$$

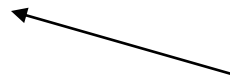
is called the *linear approximation* or *tangent plane approximation* of f at $(1, 1)$.

$$f(1.1, 0.95) \approx 4(1.1) + 2(0.95) - 3 = 3.3$$

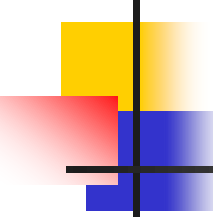
$$f(1.1, 0.95) = 2(1.1)^2 + (0.95)^2 = 3.3225.$$

$$L(2, 3) = 11$$

$$f(2, 3) = 17.$$



What does this suggest?



In general, we know from [2] that an equation of the tangent plane to the graph of a function f of two variables at the point $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

The linear function whose graph is this tangent plane, namely

$$\boxed{3} \quad L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linearization** of f at (a, b) and the approximation

$$\boxed{4} \quad f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linear approximation** or the **tangent plane approximation** of f at (a, b) .

What if f is not continuous at (a, b) ?

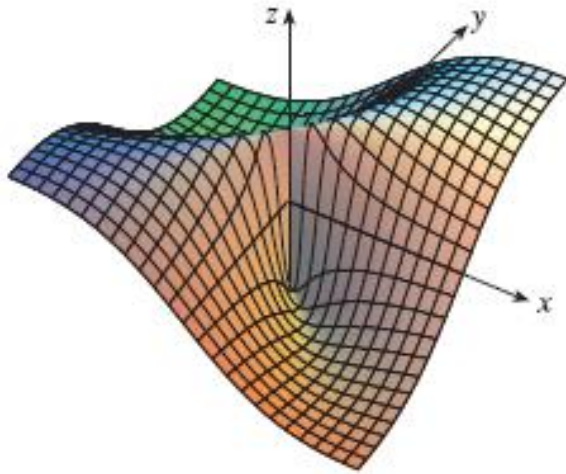


FIGURE 4

$$f(x, y) = \frac{xy}{x^2 + y^2} \text{ if } (x, y) \neq (0, 0),$$

$$f(0, 0) = 0$$

- What is the value of the first order partial derivatives at $(0,0)$?
- What about the linear approximation thereat?
- What about the value of the function along $y=x$?

7 Definition If $z = f(x, y)$, then f is **differentiable** at (a, b) if Δz can be expressed in the form

$$\Delta z = f_x(a, b) \Delta x + f_y(a, b) \Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

where ε_1 and $\varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

where $\Delta z = f(a + \Delta x, b + \Delta y) - f(a, b)$

Theorem: If f is differentiable, it is continuous

What do the following examples suggest?



Example Show that the function defined by $f(x, y) = (x^{\frac{1}{3}} + y^{\frac{1}{3}})^3$ is continuous and has partial derivatives at the origin but not differentiable thereat.

Example Show that the function defined by

$$f(x, y) = \begin{cases} x^2 \sin(\frac{1}{x}) + y^2 \sin(\frac{1}{y}) & \text{if } xy \neq 0 \\ x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0, y = 0 \\ y^2 \sin(\frac{1}{y}) & \text{if } x = 0, y \neq 0 \\ 0 & \text{if } x = y = 0 \end{cases}$$

is differentiable at the origin, but not continuously differentiable thereat because $f_x(x, y)$ is not continuous there.

Why, only at specific points we calculate partial derivatives from definition?

Definition (in V_n) $\longrightarrow$ A function $f(\mathbf{x})$ is said to be differentiable at a point $\mathbf{a}$ if

$$\lim_{\mathbf{h} \rightarrow 0} \frac{f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - \mathbf{C} \cdot \mathbf{h}}{|\mathbf{h}|} = 0$$

8 Theorem If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .



Example: Show that the function f , where

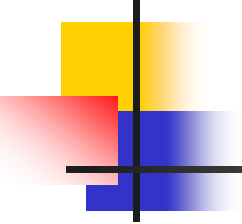
$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & \text{if } x^2 + y^2 \neq 0 \\ 0 & \text{if } x = y = 0 \end{cases}$$

is differentiable at the origin.

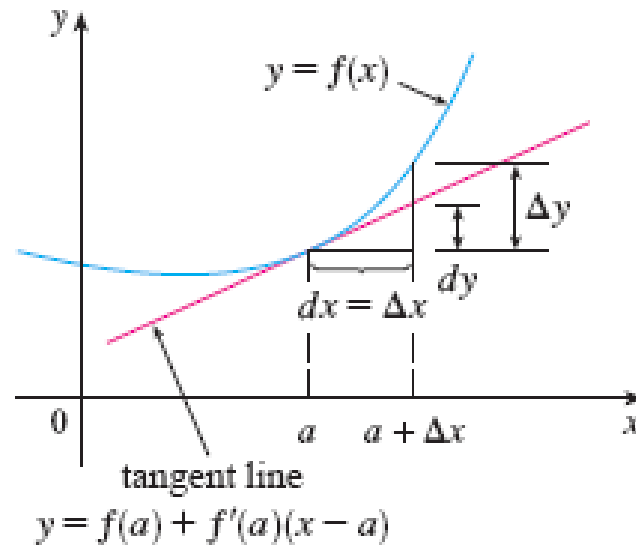
Example: Consider the function f , where

$$f(x, y) = \begin{cases} x^2 \sin \frac{1}{x} + y^2 \sin \frac{1}{y} & \text{if } xy \neq 0 \\ x^2 \sin \frac{1}{x} & \text{if } x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y} & \text{if } x = 0, y \neq 0 \\ 0, & \text{if } x = y = 0 \end{cases}$$

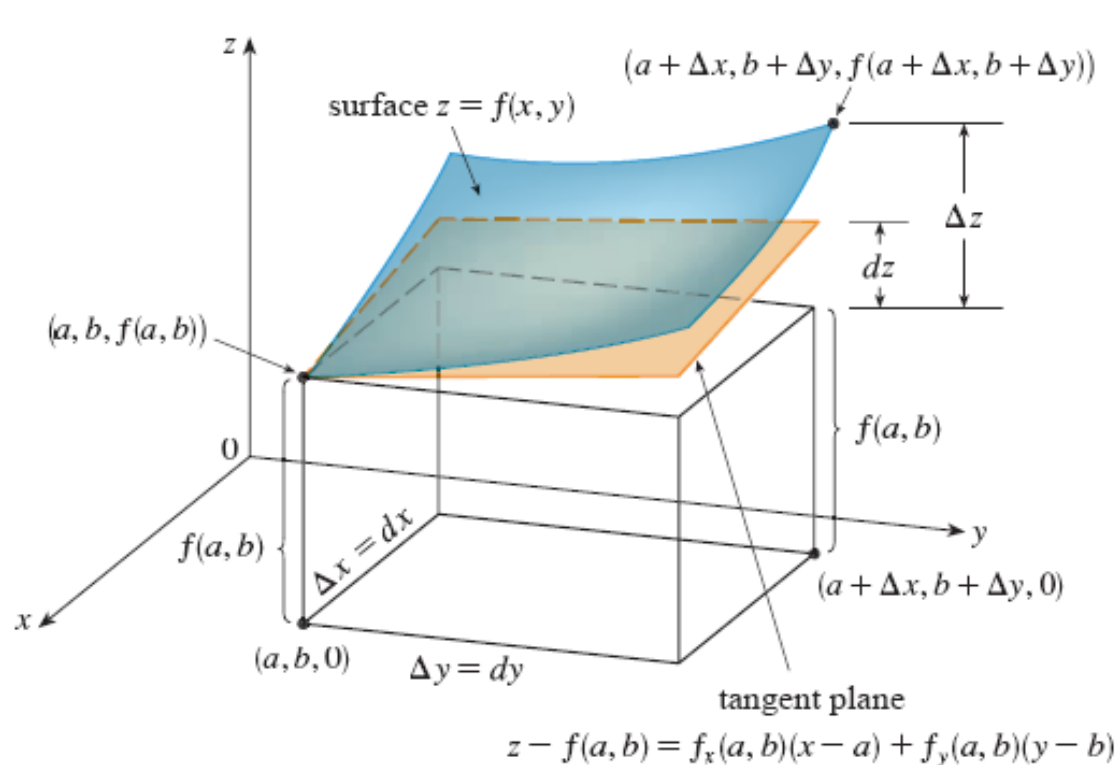
Is the function continuously differentiable at the origin? What about its differentiability thereat?

- 
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- ✚ If f is continuously differentiable, then f is differentiable
 - ✚ If f is differentiable, then all partial derivatives of f exist
 - ✚ If f is differentiable, it is continuous

Differential: Function of one variable



Differential: Function of two variables



$$dz = f_x(x, y) dx + f_y(x, y) dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$